

Double Pendulum and The Chaos Theory

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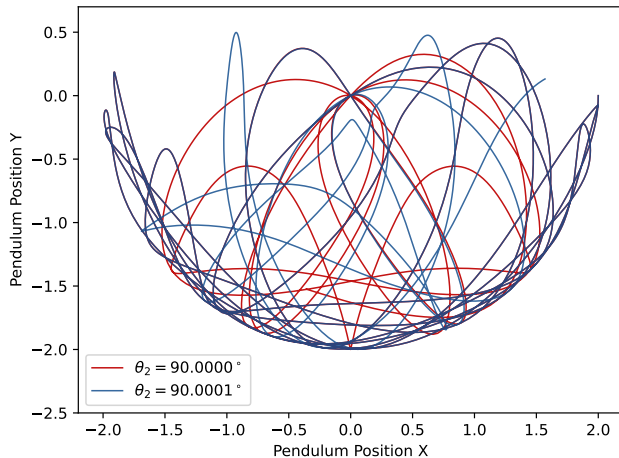
Related CALCAB Knowledge

Differential Equation: It is the core of simulation algorithm we applied in this project, the entire simulation was run based on a differential equation derived by *Lagrangian Mechanics*, so the differential equation we learned in unit 7 is the core of our project.

Limit and Approaching: In our project, the observation of the simulation accuracy as time step changes is also a core part. By speculating the result approaches a value as step decreases (becomes more accurate), we could choose a suitable time step for simulation that is both efficient and accurate.

Introduction To The Chaos Theory

Two Slightly Different Pendula Simulation Outcome



Uncertain Outcome: In a chaotic system described in the Chaos Theory, even a *negligible* disturbance or initial differences could cause an incredibly different result.

Determinable though Chaotic: Chaotic doesn't mean random and indeterminate, instead, chaotic systems could be predicted by calculations and modeling if one has incredibly precise data, though the result is input-sensitive.

Double Pendulum, Where Newtonian Physics Dies.

Introduction To DE Calculation Approaches

**Fourth-Order Runge-Kutta
(RK4) Approach:**